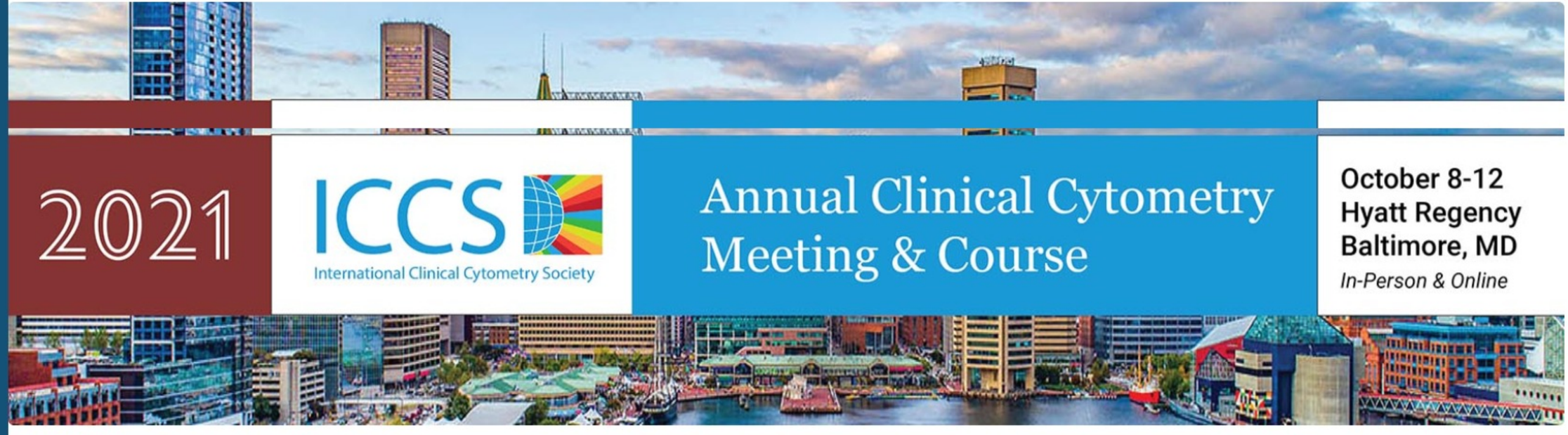

Abstract Preview

The following is a preview of your submission abstract #149:

Artificial Intelligence-Assisted Classification Tool for Minimal Residual Disease Assessment by Flow Cytometry in Patients with Acute Myeloid Leukemia

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Background: Flow cytometric data analysis for MRD detection in AML is complex, time-consuming and requires significant analytic expertise. This study aimed to improve previously developed AI tools thatcould automatically and efficientlydetect AML MRD. **Methods**: 209 bone marrow samples from patients without AML (n=107) or with residual AML [n=102; stratified in 4 tiers, based on the level of residual disease (RD)] were acquired on BD FACSLyric cytometers. Uncompensated FC data was used to develop algorithms for AML RD detection and RD tier classification, based on the "ground truth" confirmed by hematopathologic evaluation. We leveraged the Gaussian Mixture Model and Fisher vectorization (GMM-FV) approach to characterize data distribution. Support Vector Machine (SVM) was applied to separate AML RD "negative" and "positive" samples by a hyperplane. Random Forest (RF) was applied on multiclass prediction for RD tier classification. **Results**: AML RD detection and RD Tier Classification models achieved performance of 0.913 and 0.919 AUC, respectively. Both algorithms achieved >0.9 specificity in identifying samples with >0.1% residual AML (**Table 1**). The 3D visualization approaches demonstrated distinct separation between AML RDP 0% and those with RDP >0% (**Figure 1a)**. According to the confusion matrix analysis of RDP Tier Classification model (**Figure 1b)**, 35.3% of mis-classified cases showed <0.1% RD. **Conclusion:** This study demonstrates the potential of machine learning algorithms to support the detection of residual AML, at clinically relevant MRD cut-offs, as recommended by current practice guidelines. Additional multi-institution studies are needed to further optimize and validate these models for future clinical use.

Table 1. AML and RD prediction model performance.

Model	Accuracy*	AUC	UAR	Specificity **	Sensitivity **	PPV**	NPV**
AML(+/-)	0.880	0.913	0.880	0.907	0.853	0.897	0.866
Residual disease (negative, <0.1%, >= 0.1% & <1%, >=1% & <10%, >=10%)	0.837	0.919	0.752	0.975	0.920	0.964	0.944

* Concordance rate between algorithm prediction to the "ground truth" confirmed by integrated hematopathologic evaluation.
** RD 0.1% as cut off.

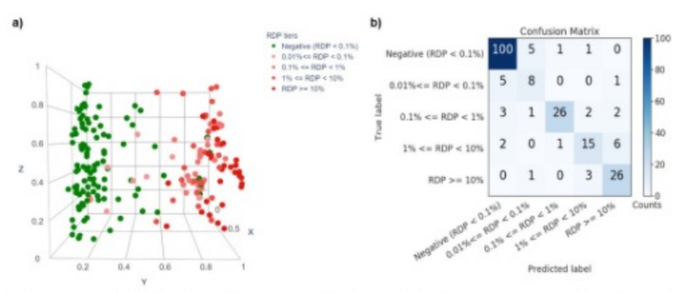

Figure 1. Visualization of AML RD Binary Classification Model and Concordance Comparison of AML RD Tier Classification Model.

a) Visualization plotted via the first 2 principle components (X- and Z-axis) of the principle components analysis of fisher vector, and prediction probability (Y-axis) of AML RD positive, using the GMM and SVM parameters of the fold with highest accuracy. Each dot represents an individual specimen and different color indicate different residual disease group.

b) Confusion matrix analysis is used for comparing algorithm prediction concordance with ground truth in each AML RDP subgroups of the training set